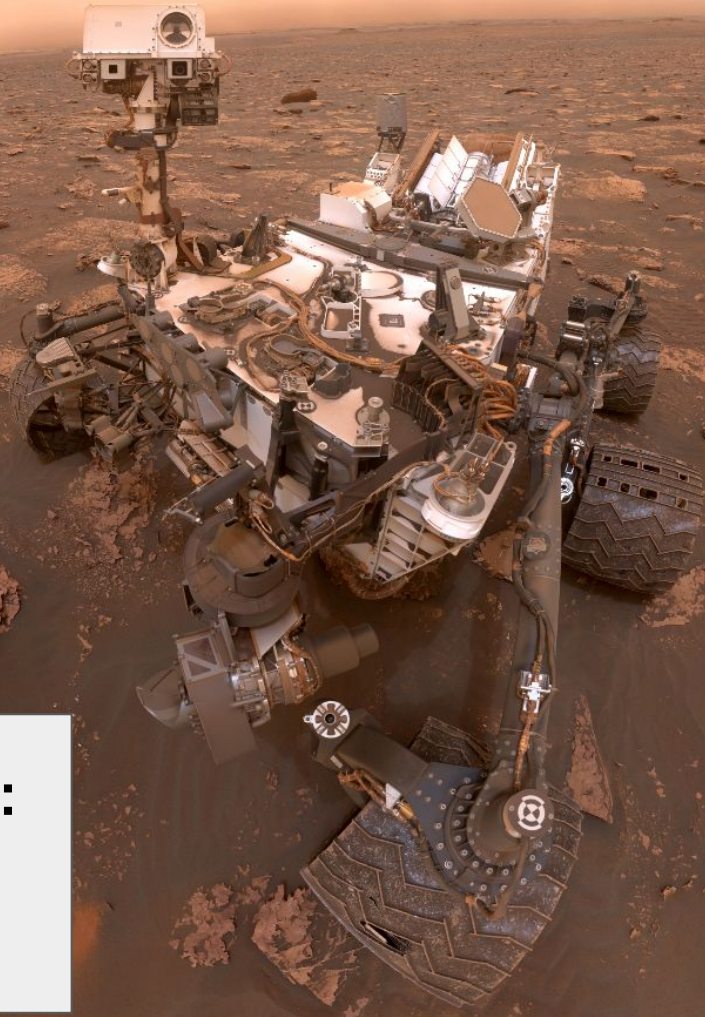


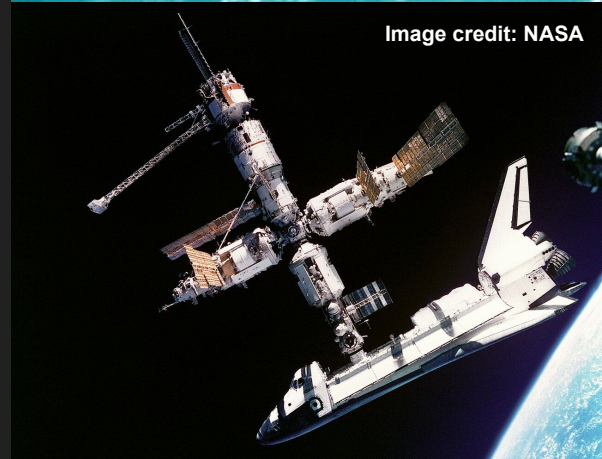
Lesson 4: Rovers and probes

**NASA and Space Exploration:
A Journey through the
Solar System**



Recap from Last Week

- Several space stations have been built
 - Skylab was one of the first designs
 - MIR was the first multi-module system
 - The ISS is a collaboration between USA, Russia, Japan, Europe, and Canada
- On board the ISS, many experiments are conducted
 - Strange behaviour of honey in space
- Living in space is not easy
 - You saw videos on how to sleep in space and how to use the toilet in space!



Recap from Last Week

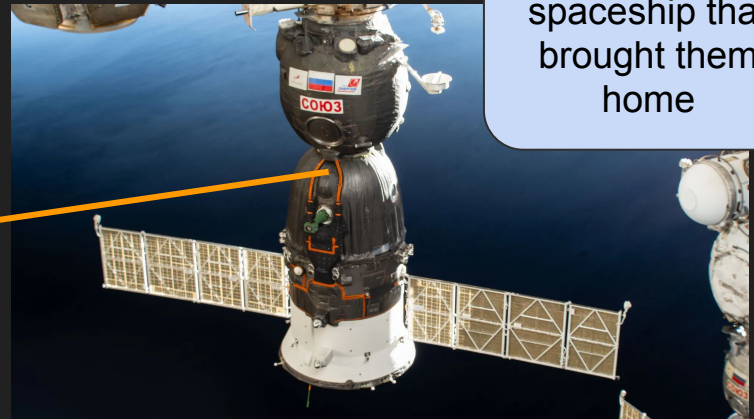
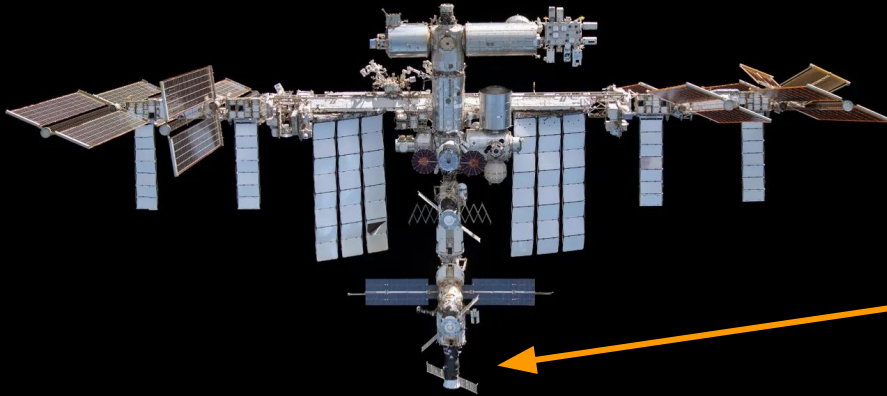
- Who has spotted the ISS overhead at night?

Should have looked something like this:



Space News!

- NASA astronaut Chris Williams and Roscosmos cosmonauts Sergey Kud-Sverchkov and Sergei Mikaev left the ISS and came back to Earth
- It took them 4 hours to go from the ISS to Earth



This was the
spaceship that
brought them
home

Space News!



NASA ASTRONAUT CHRIS WILLIAMS

SOYUZ MS-28 RE-ENTRY & LANDING

Today's Topic: Rovers and Probes



A map of all Solar System Probes

The sun
has
been
visited
22 times

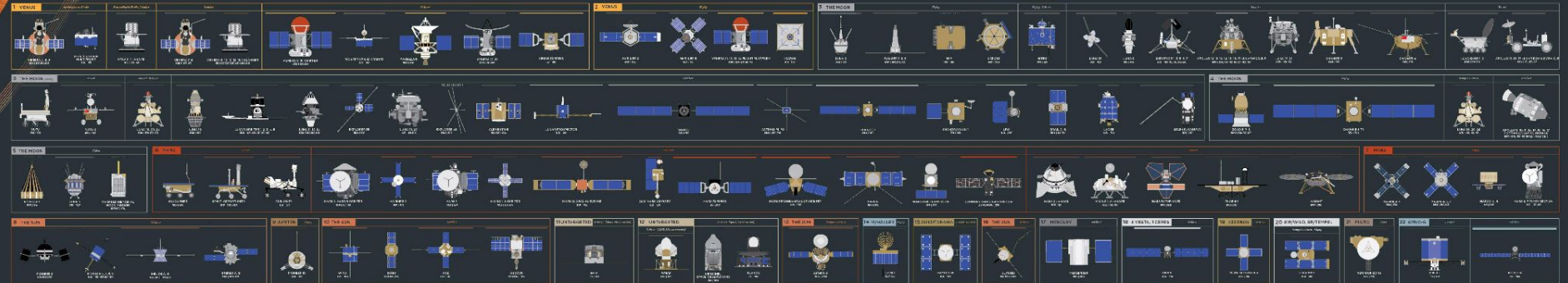
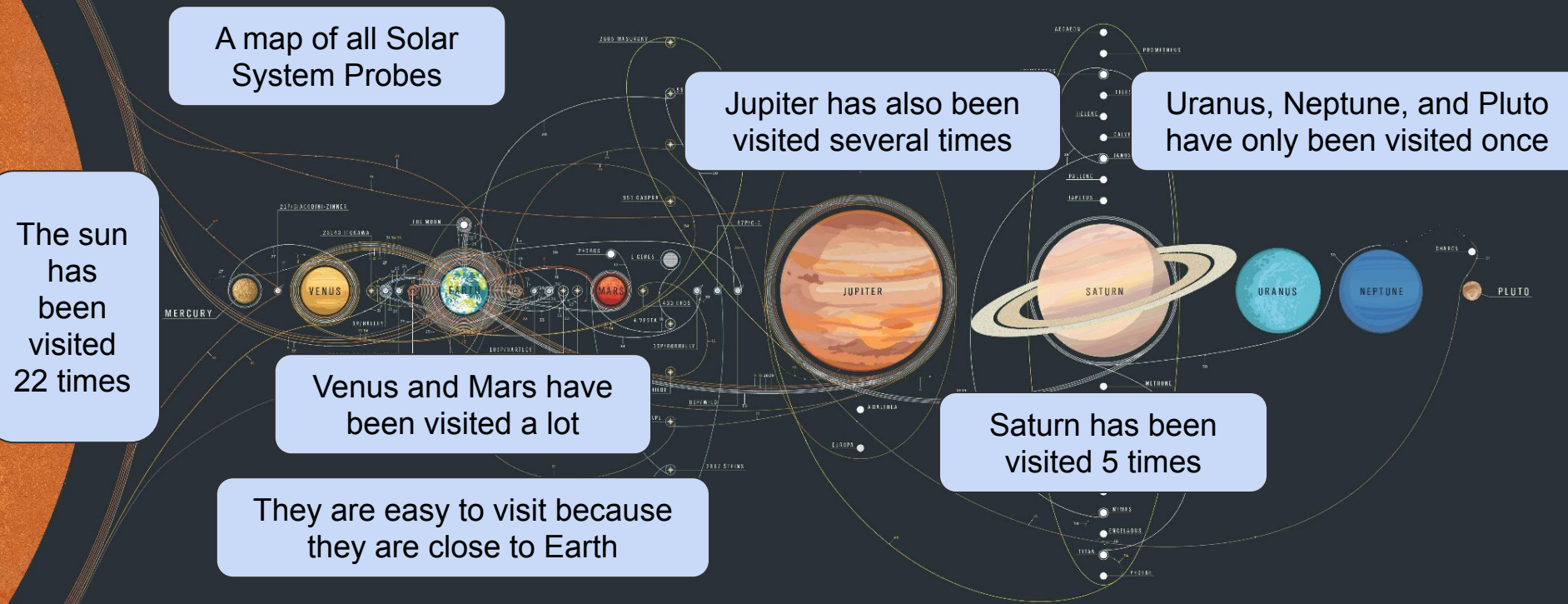
Venus and Mars have been visited a lot

They are easy to visit because they are close to Earth

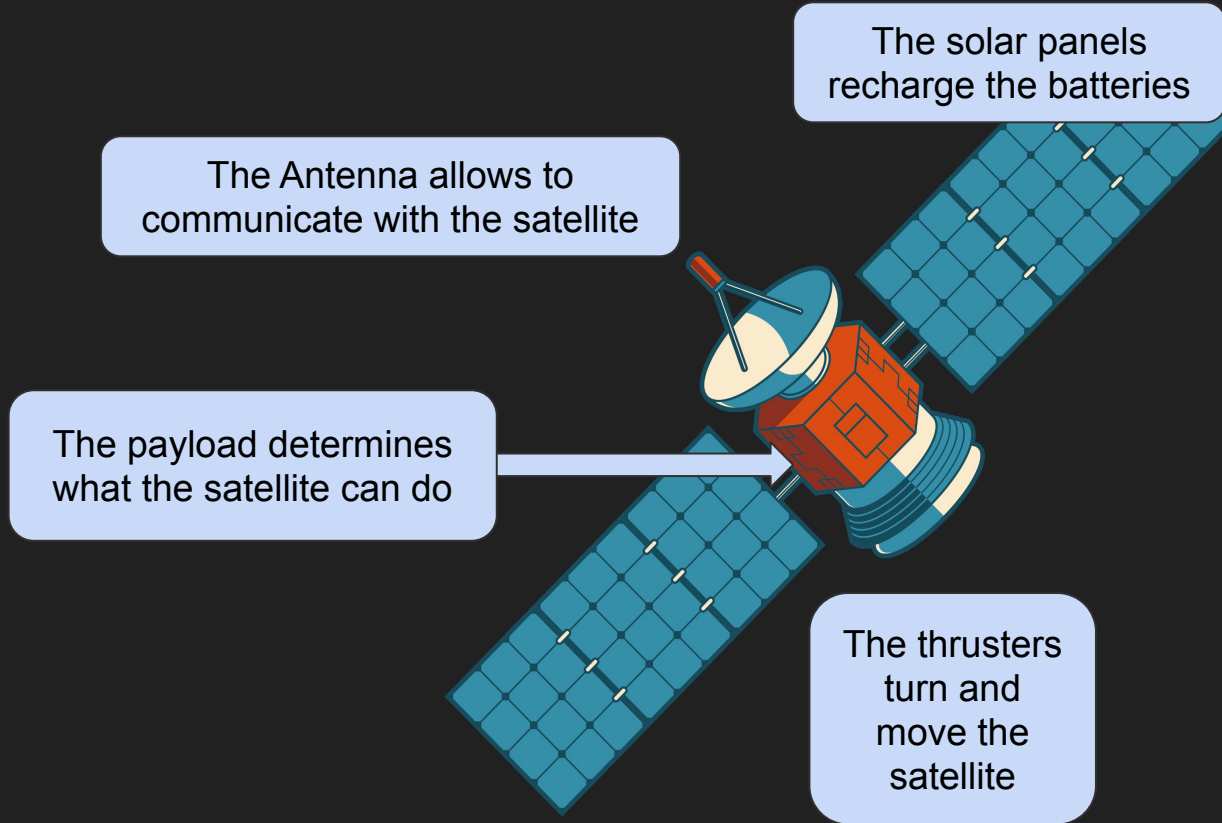
Jupiter has also been visited several times

Uranus, Neptune, and Pluto have only been visited once

Saturn has been visited 5 times



Important parts of a Satellite Probes



Satellites reach space the same way humans do.
On Rockets!



Image credit: France-Presse — Getty Images

Which planet is this?

Venus

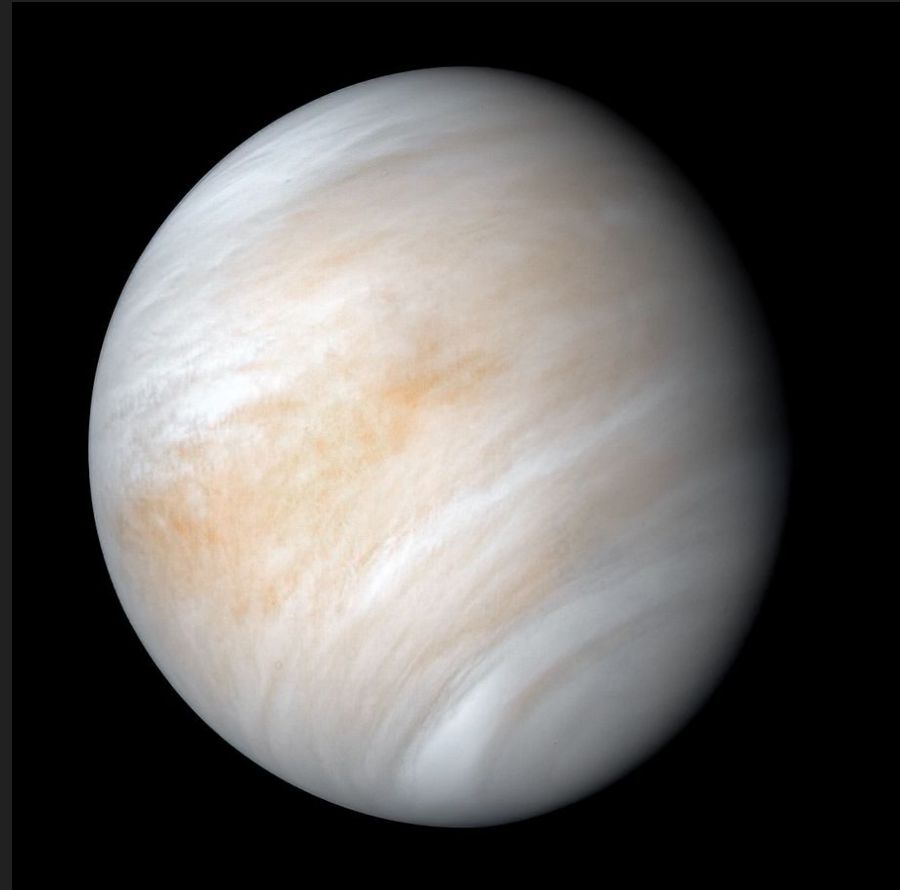
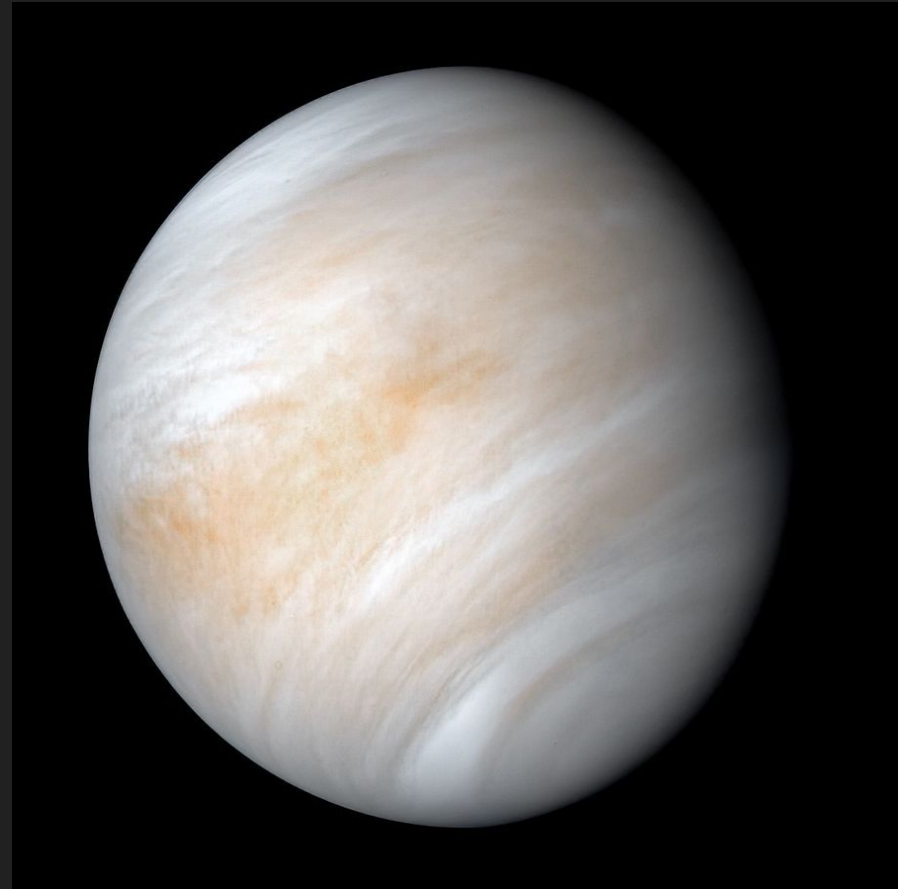


Image credit: NASA

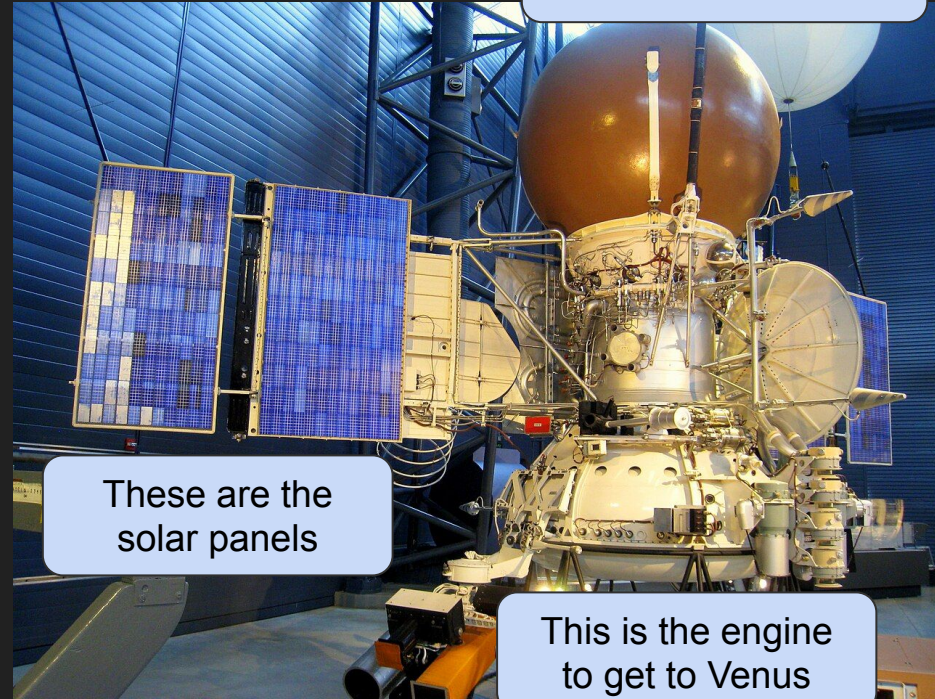
Venus

- Venus has a thick atmosphere made of CO₂ (carbon dioxide)
- Before we sent probes, we did not know anything about its surface and very little about its atmosphere.
- It was the first planet we visited.
- Venus has been visited 39 times.
- The most famous programm visiting Venus is the Venera programm.



The Venera Probes on Venus

- Within the Venera program of the Soviet Union:
 - 18 probes were launched
 - 13 probes entered Venus
 - 10 reached the surface
- The probes survived between 23 minutes to 2 hours
- They collected valuable information about Venus' atmosphere and surface



This is the lander that entered Venus

These are the solar panels

This is the engine to get to Venus

The Venera Probes - Highlights

- Venera 7 reached Venus' surface. (15 December 1970)
- Venera 9 sent the first picture of the surface of another planet. (8 June 1975)



- Venera 13 sent the first recorded sound from another planet. (1 March 1982)

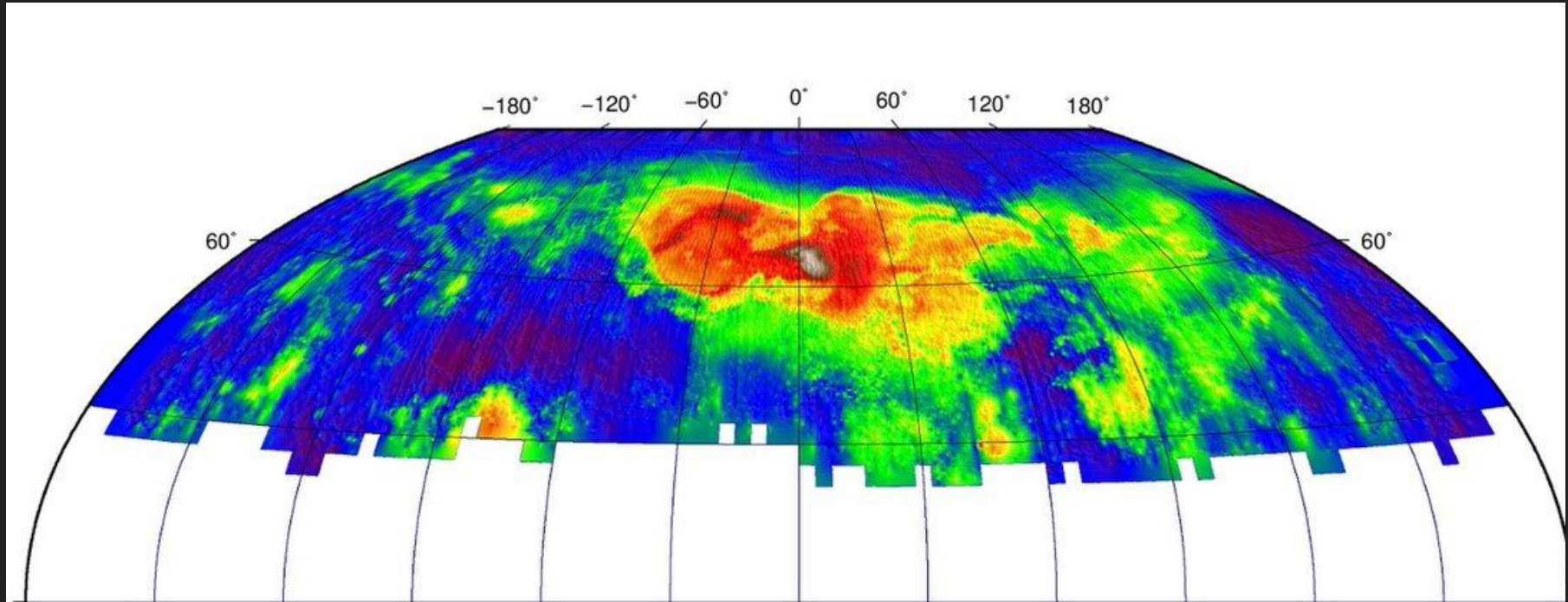


What you can hear are winds!

This recording proved that Venus also has winds.

- Venera 15 measured the mountains and valleys of Venus. (10 October 1983)
 - The red parts are mountains, the blue parts are valleys

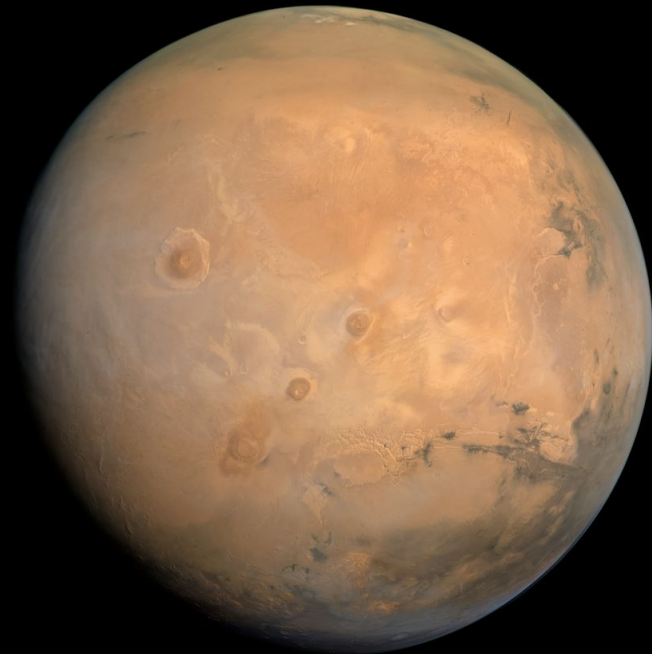
The Venera Probes - Highlights



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Which planet is this?

Mars

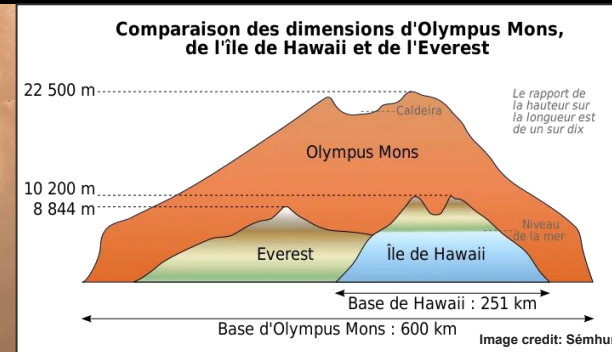
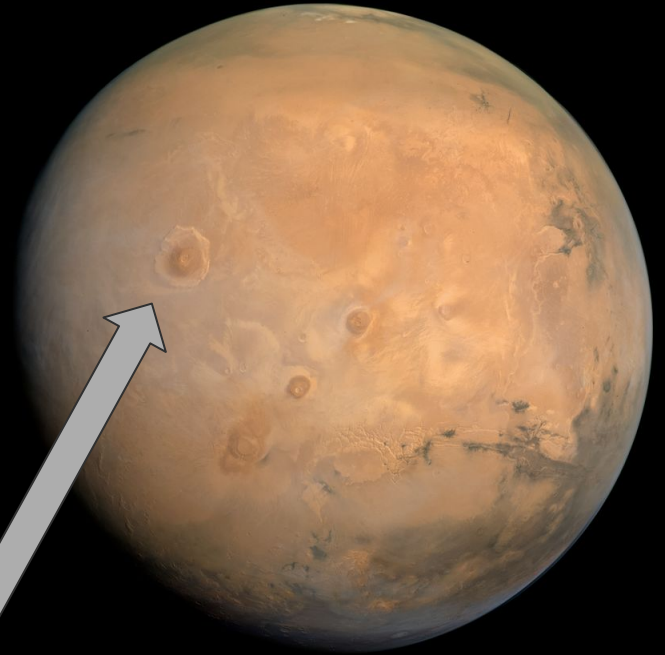


Mars

- Mars is our closest neighbour.
- It has a very thin atmosphere.
- Mars has been visited 35 times.
- There are 10 active satellites and 2 active rovers on Mars right now.
- **Fun Fact:** Mars has the highest mountain in the Solar system! - Olympus Mons

What is it named after?

The Olymp is where the greeks god live!



Important parts of a Mars Rover

Nuclear battery that powers the rover even during sand storms

A camera to send us pictures from Mars and to avoid obstacles

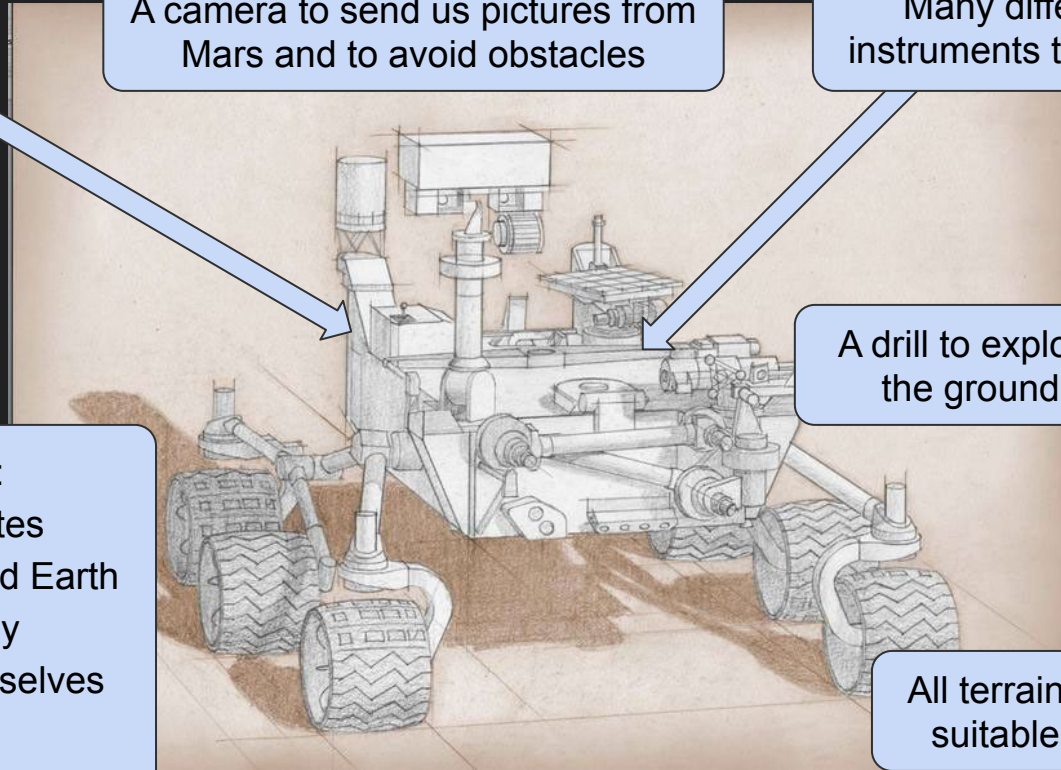
Many different science instruments to measure Mars

A drill to explore the ground

Fun Facts about Mars rovers:

- There is a 3 to 20 minutes delay between Mars and Earth
- Rovers are not driven by Humans but drive themselves
- And

All terrain wheels which are suitable to drive on Mars





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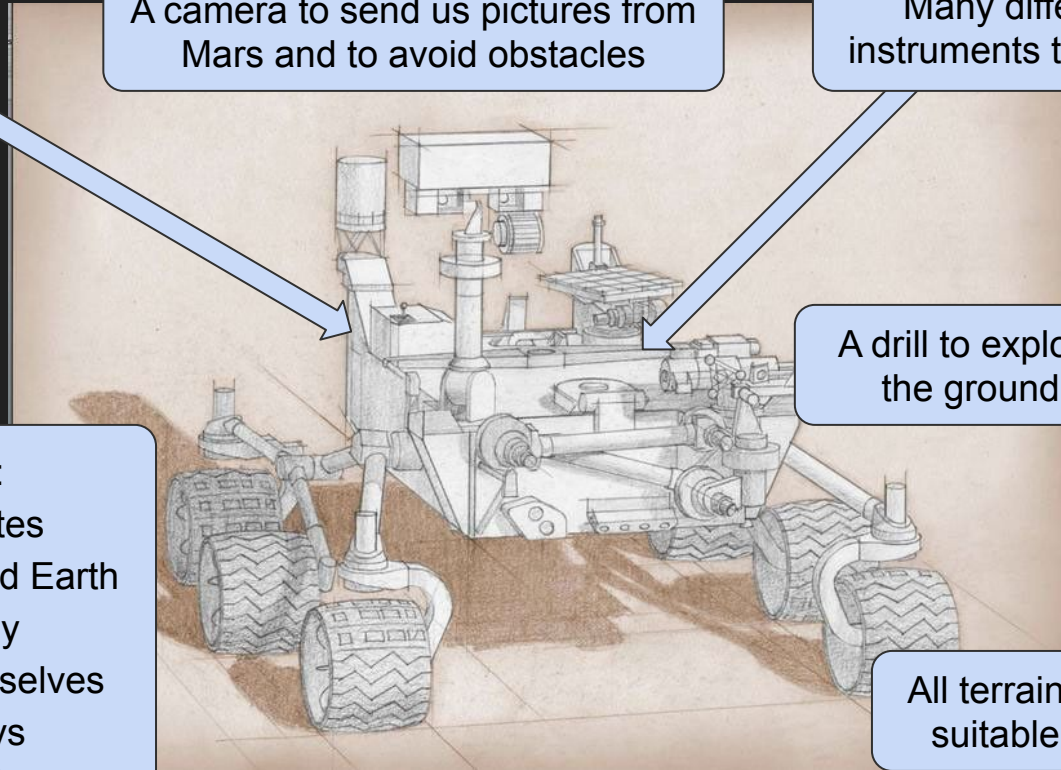
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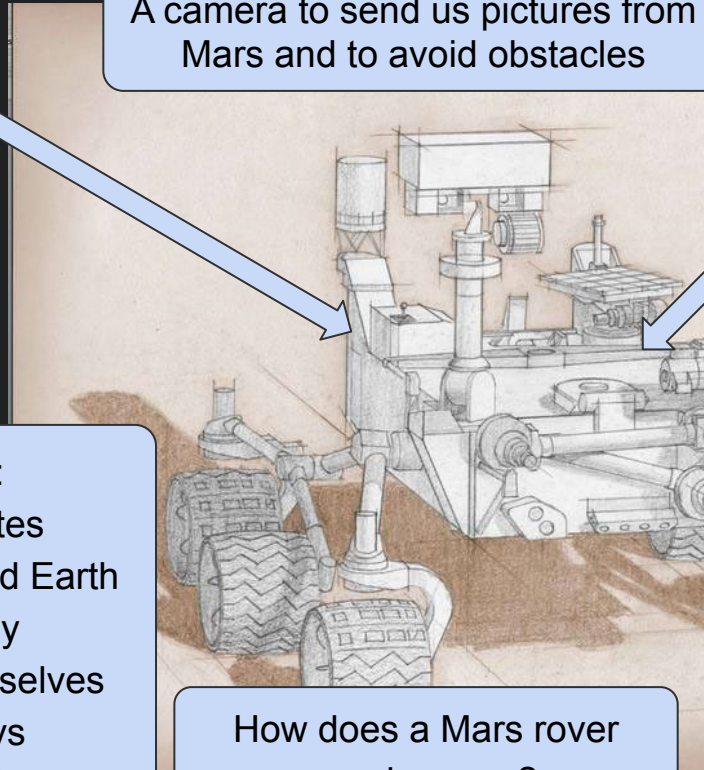
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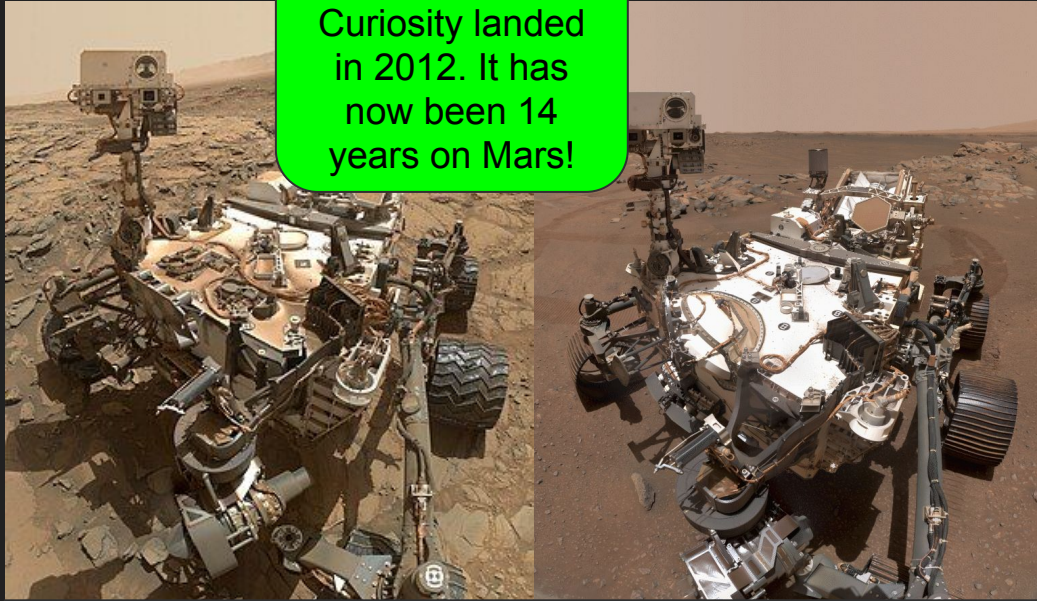
How does a Mars rover reach space?



Mars Rovers - A Family Picture

Still active:

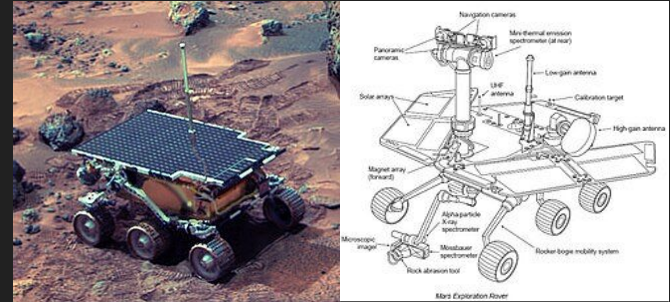
Fun Fact:
Curiosity landed
in 2012. It has
now been 14
years on Mars!



NASA Curiosity

NASA Perseverance

Past missions:



NASA Sojourner

NASA Sojourner



NASA Opportunity

CNSA Zhurong

What have we learned from the Rovers?



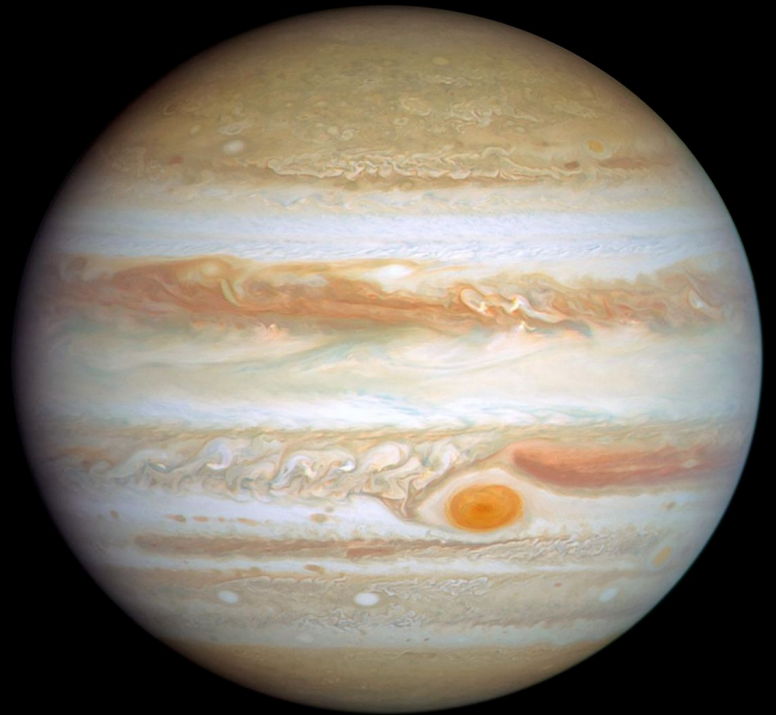
What have we learned from the Rovers?

- We learn about the surface and geology of mars.
- We found
 - Mountains and craters from meteorites impacts
 - We found dried up lake grounds and riverbeds
- One of my favourites: They found a honeycomb pattern that looks similar to the salt flats in Bolivia.



Which planet is this?

Jupiter



Jupiter

- Jupiter is the biggest planet in the Solar System.
- It is made mostly out of gas and we don't know if it has a surface.
- Jupiter has been visited 13 times.
- Jupiter's moon Ganymed was also visited once.



Jupiter Exploration

- 7 spacecrafts visited Jupiter but just briefly
- 2 spacecrafts went to Jupiter and stayed there (Juno and Galileo)
- **Fun Fact:** In Roman mythology, Juno is Jupiter's wife!



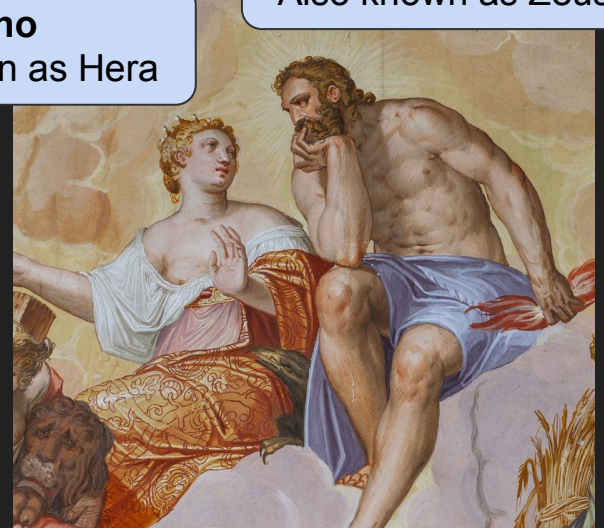
The spacecraft
Juno



The spacecraft
Galileo

Juno
Also known as Hera

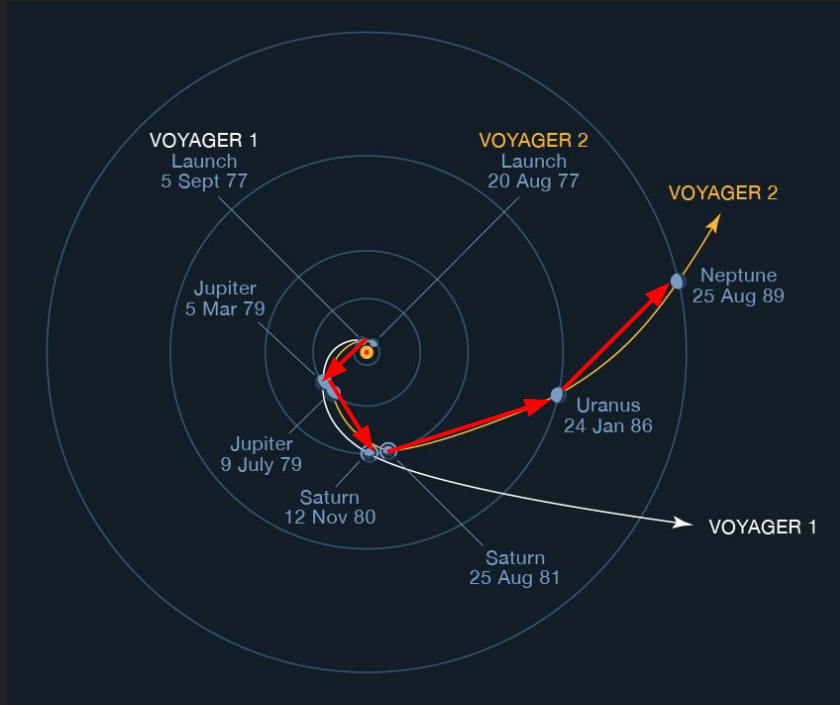
Jupiter
Also known as Zeus



How do we reach Jupiter?



How do we reach Jupiter?

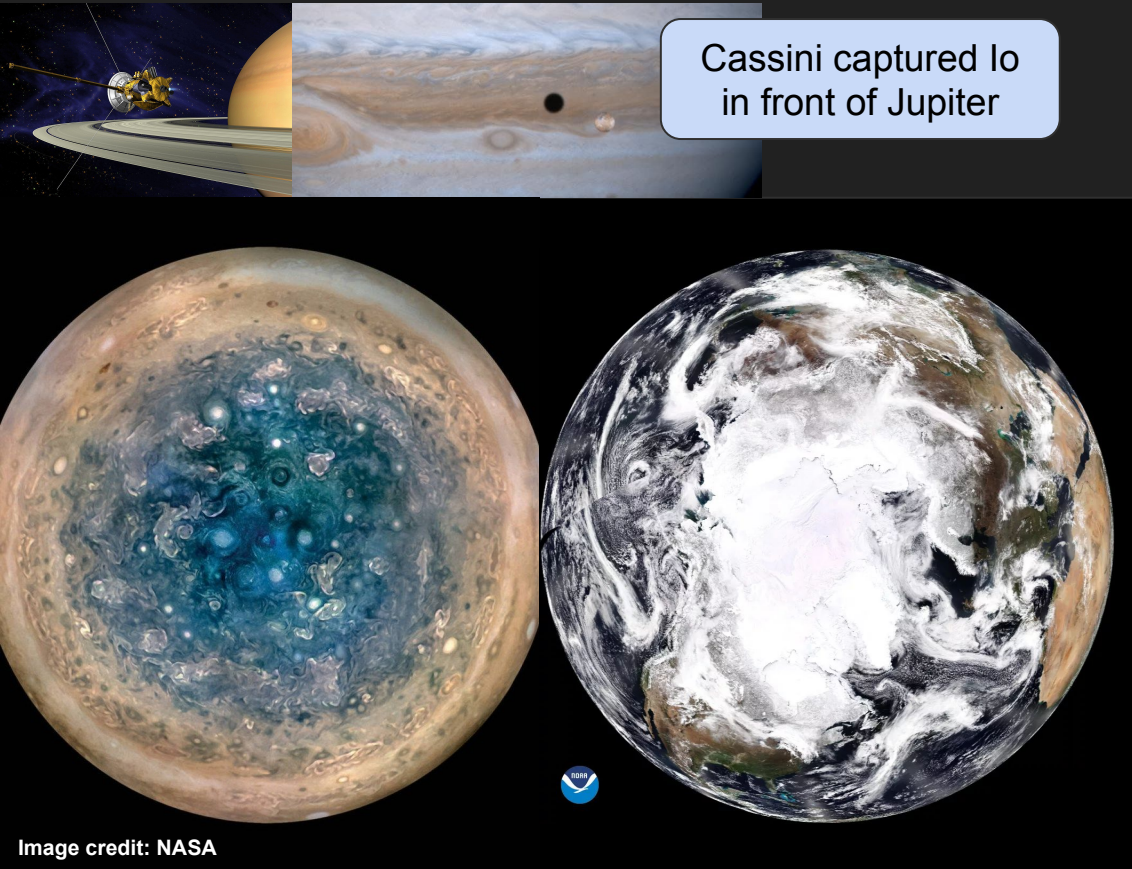


Flying close to other planets gives a push.
That way, the satellite can travel even further outwards:

1. Jupiter
2. Saturn
3. Uranus
4. Neptune

Voyager 1 & 2 are leaving the Solar System and will never return.

Jupiter Exploration - What did we learn?

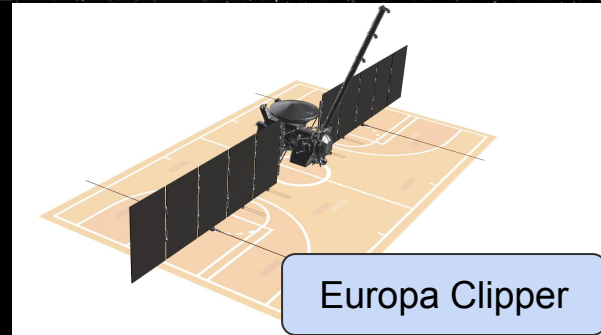


Cassini captured Io
in front of Jupiter

- Juno gave us detailed images of the poles of Jupiter
- The poles of Jupiter looks very different than the rest
- Earth's poles also look very different than the rest
- Thanks to Juno we can now study the weather on Jupiter!
- Juno had 9 other cameras

Jupiter's Moons

- Jupiter's moons Ganymede and Europa might have liquid oceans underneath a thick layer of ice
- Europa clipper is currently flying to Europa (arrival 11 April 2030)
- JUICE is currently flying to Ganymede (arrival December 2034)



Which planet are these?

What about this fellow?

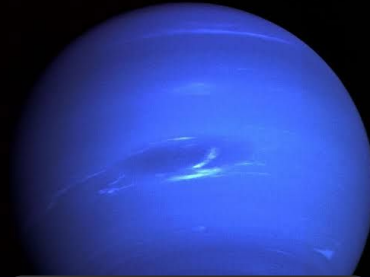
Saturn



Pluto



Neptune

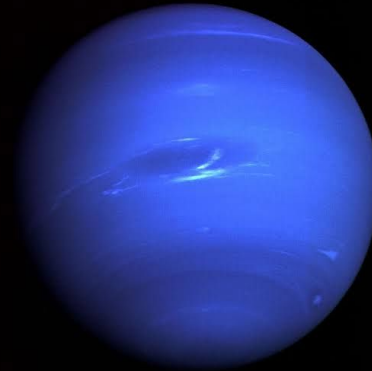


Uranus



Saturn, Neptune, Uranus, and Pluto

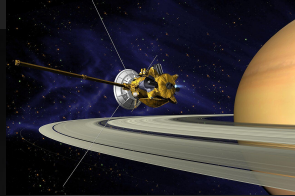
- Saturn, Neptune, and Uranus are all mostly made from gas.
- Pluto is no longer considered a Planet. It is a dwarf planet.
- In total, there were a total of only 8 missions in total to Saturn, Neptune, Uranus, and Pluto.



Saturn's Moons

- Enceladus is the 6th biggest moon of Saturn. Water plumes were detected!
- Titan is Saturn's largest Moon with a methane atmosphere.
- When Cassini flew by, it dropped Huygens onto its surface.
- This gave us the first images of the atmosphere of a Moon!

Water plumes of Enceladus

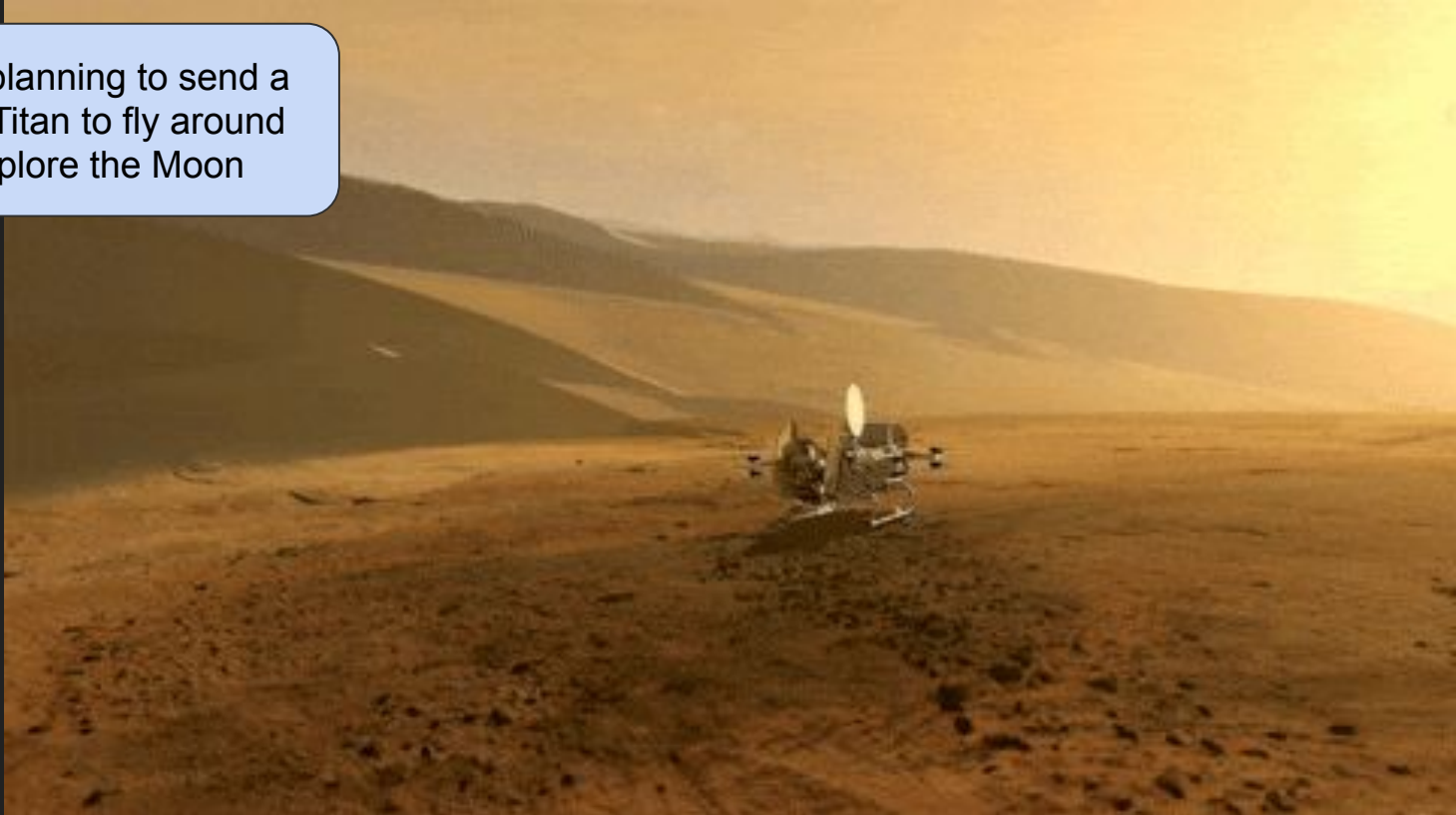


Titan's surface imaged by the Huygens probe



Cool Future Mission: Titan Dragonfly

NASA is planning to send a drone to Titan to fly around and explore the Moon



Today's Activity: RC Rover

→ We will use an RC car to test the challenges of Mars rovers